A Proof of the Tratnik–Ye Medianity Conjecture for Resonance Graphs of Perfect Matchings on Surfaces*

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Abstract

Tratnik and Ye conjectured that if G is a graph embedded in a closed surface and $F \subsetneq F(G)$ is a proper set of even faces, then every connected component of the resonance graph $R(G;F)$ is a median graph. This conjecture extends the medianity phenomena known for resonance graphs in planar and chemical-graph settings to graphs embedded on arbitrary closed surfaces. We prove the conjecture.

The proof does not rely on a distributive-lattice structure or on an a priori isometric embedding into a hypercube. Instead, for each connected component H of $R(G;F)$, we construct a cubical complex X_H whose 1-skeleton is exactly H. The cubes of X_H are generated by collections of pairwise boundary-disjoint facial rotations, that is, by rotations that can be performed independently. The vertex links of X_H are flag, so by Gromov's criterion it remains to prove simple connectedness.

The main difficulty is global. We prove simple connectedness by reducing closed words in facial rotations. The key local tool is a parity-propagation lemma: if an even face f is alternating before and after a word of facial rotations in which f does not occur, then the occurrence parities of all face sectors incident with ∂f, except the sector of f, are forced to be equal. Applied to a shortest non-null-homotopic loop, this parity constraint produces an infinite strictly descending chain of nonempty subwords, which is impossible. Hence X_H is simply connected, therefore CAT(0), and its 1-skeleton H is a median graph.

This proves the Tratnik–Ye medianity conjecture and gives an intrinsic nonpositive-curvature explanation for the median structure of resonance graph components on surfaces.

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1 Introduction

Resonance graphs encode the structure of perfect matchings under local transformations. They have long been studied in chemical graph theory, especially in connection with Kekulé structures, benzenoid systems and fullerene-type graphs. In many planar settings, resonance graphs are known to have strong metric and order-theoretic properties; in particular, they often arise as median graphs through distributive-lattice constructions.

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The purpose of this paper is to prove a conjecture of Tratnik and Ye which asks whether such medianity persists for resonance graphs of perfect matchings on closed surfaces. Their conjecture goes beyond the classical planar setting and removes the direct availability of the usual lattice machinery. The main point of the present work is to show that the correct underlying structure is not necessarily a lattice, but a CAT(0) cube complex built from independent facial rotations.

Let G be a finite graph embedded in a connected closed surface Σ, and let $F(G)$ denote the set of faces of the embedding. We use the standard strong-embedding assumptions used by Tratnik and Ye [6]: the relevant faces are bounded by facial cycles, and the face-dual is connected. Let $F \subseteq F(G)$ be a set of even faces. The resonance graph $R(G; F)$ is defined as follows. Its vertices are the perfect matchings of G. Two perfect matchings M and N are adjacent if and only if

$$M \triangle N = E(f)$$

for some $f \in F$. Equivalently, N is obtained from M by flipping the M-alternating facial cycle ∂f.

Tratnik and Ye proved that when $F \subsetneq F(G)$, every connected component of $R(G; F)$ is an induced subgraph of a hypercube [6, Theorem 3.5]. They also observed that this embedding need not be isometric, and therefore it does not by itself imply that the component is median. They formulated the following conjecture [6, Conjecture 5.3].

Theorem 1.1 (Tratnik–Ye medianity conjecture)**.** *Let G be a graph embedded in a closed surface, and let $F \subsetneq F(G)$ be a set of even faces. Then every connected component of $R(G; F)$ is a median graph.*

Recall that a connected graph is median if, for every three vertices x, y, z, there is a unique vertex lying simultaneously on a shortest x-y path, a shortest y-z path and a shortest z-x path. Median graphs are precisely the 1-skeleta of CAT(0) cube complexes. Thus the conjecture naturally suggests looking for a cubical structure behind the resonance graph.

We prove the conjecture by constructing such a structure directly. Let H be a connected component of $R(G; F)$. We define a cubical complex X_H whose vertices and edges are exactly the vertices and edges of H. Higher-dimensional cubes are attached whenever several facial rotations can be performed independently. More precisely, if $S = \{f_1, \ldots, f_k\} \subseteq F$ is a set of pairwise boundary-disjoint faces that are alternating with respect to a perfect matching M, then flipping any subcollection of S produces a Boolean k-cube of perfect matchings. These Boolean cubes are precisely the cubes attached in X_H.

This construction is intrinsic: it uses only the local commutation of independent facial rotations. It is not based on choosing a particular hypercube embedding. The link of each vertex of X_H records which facial rotations are available at the corresponding perfect matching. Since pairwise boundary-disjoint rotations can be performed simultaneously, every clique in a vertex link is filled by a simplex. Hence all vertex links are flag.

By Gromov's link criterion, a cube complex is CAT(0) if and only if it is simply connected and all vertex links are flag. Since the flag condition follows from the construction, the essential part of the proof is to show that X_H is simply connected.

The proof of simple connectedness is based on closed words in facial rotations. A closed path in H gives a word

$$w = f_1 f_2 \cdots f_t$$

in face labels, where each letter records one facial flip. There are two elementary homotopy moves: deleting a consecutive pair ff, corresponding to flipping a face and immediately flipping it back, and interchanging consecutive letters $fg \leftrightarrow gf$ when the facial cycles ∂f and ∂g are boundary-disjoint, corresponding to a square in X_H.

The global parity condition comes from the assumption $F \subsetneq F(G)$. If a face word is closed, then every face of G occurs an even number of times. Indeed, the total symmetric difference of the facial boundaries appearing in the word is empty, and connectedness of the face-dual propagates parity from face to face. Since some face outside F never appears, all faces have even occurrence parity. This parity fact is the starting point for the reduction argument.

The key local ingredient is a parity-propagation lemma. Suppose an even face f is alternating at both ends of a path whose face word does not contain f. Then, at each vertex of ∂f, the occurrence parities of all face sectors other than the sector of f are forced to be equal. Consequently, all faces meeting f have the same parity in that word. This local constraint is what prevents a shortest nontrivial closed word from existing.

Indeed, assume that there is a shortest closed edge path in H that is not null-homotopic in X_H. Its face word is fully reduced: after using square commutations, no immediate cancellation ff can occur. Choose a face p_0 appearing in this word and an adjacent face p_{-1} not appearing in it; such a pair exists because $F \subsetneq F(G)$ and the face-dual is connected. Between two consecutive occurrences of p_0, we obtain a nonempty subword U_0. The parity-propagation lemma forces every face meeting p_0 to have even parity inside U_0. Full reducedness then guarantees that some face p_1 meeting p_0 occurs inside U_0, and because its parity is even it occurs at least twice. Taking the subword between two consecutive occurrences of p_1 produces a strictly smaller nonempty word U_1. Repeating the argument yields an infinite strictly descending chain

$$U_0 \supsetneq U_1 \supsetneq U_2 \supsetneq \cdots$$

of nonempty finite words, which is impossible. Thus every closed edge path is null-homotopic, and X_H is simply connected.

It follows that X_H is a CAT(0) cube complex. Since the 1-skeleton of a CAT(0) cube complex is a median graph, and since $X_H^{(1)} = H$, the connected component H is median. This proves the Tratnik–Ye conjecture.

The paper is organized as follows. Section 2 recalls closed rotation words and proves the basic parity lemma for closed walks. Section 3 constructs the cubical complex X_H and proves that its vertex links are flag. Section 4 establishes the local parity-propagation lemma. Section 5 proves simple connectedness by the descending-word argument. Section 6 completes the proof of the conjecture and records final remarks on the role of the assumption $F \subsetneq F(G)$.

Notation and terminology

For reference, we spell out the notation used throughout the proof.

Σ — A connected closed surface in which the finite graph G is embedded.

$V(G), E(G), F(G)$ — The vertex set, edge set, and face set of the embedded graph G.

$\partial f, V(\partial f), E(f)$ — For a face $f \in F(G)$, ∂f is its facial boundary cycle, $V(\partial f)$ is the set of vertices on this cycle, and $E(f)$ is the set of boundary edges of this cycle.

F — A fixed proper subset of $F(G)$ consisting of even faces. The condition $F \subsetneq F(G)$ is used essentially in Lemma 2.1 and Theorem 5.2.

M, N — Perfect matchings of G, viewed as subsets of $E(G)$. A face f is M-alternating if, along the cycle ∂f, the edges alternate between belonging to M and not belonging to M.

$\triangle$	Symmetric difference of sets. Thus $M \triangle N$ is the set of edges belonging to exactly one of M, N, and $M \triangle E(f)$ is obtained from M by toggling the boundary edges of f. The symbol $\triangle_{f \in T} E(f)$ denotes the iterated symmetric difference of the edge sets $E(f)$ over $f \in T$.
$R(G; F)$	The resonance graph whose vertices are the perfect matchings of G and whose edges are the allowed facial rotations along faces in F.
face-dual	The graph whose vertices are the faces in $F(G)$ and whose edges join two faces sharing a boundary edge. We use connectedness of this graph in the parity arguments.
face sector	At a vertex of the embedded graph, a face sector is the sector between two consecutive incident half-edges in the cyclic order determined by the embedding.
$M \xrightarrow{f} N$	A labeled step in $R(G; F)$: it means that $f \in F$ is M-alternating and $N = M \triangle E(f)$. The arrow records the direction in which a path is read and the face label of the step; it does not impose an orientation on the resonance graph.
H	A connected component of $R(G; F)$.
X_H **and** $X_H^{(1)}$	The cubical complex built from the component H in Section 3, and its 1-skeleton. For any cube complex X, the 1-skeleton $X^{(1)}$ is the ordinary graph obtained by keeping all vertices and edges of X and forgetting all cubes of dimension at least 2. We will have $X_H^{(1)} = H$.
$\mathrm{Lk}_X(x)$	The link of a vertex x in a cube complex X. Its vertices are germs of edges of X issuing from x; an edge germ is the local direction of an edge starting at x. A set of such vertices spans a simplex exactly when those edge germs lie in a common cube of X. In particular, $\mathrm{Lk}_{X_H}(M)$ denotes the link of the vertex M in X_H.
$\varepsilon_u(h)$	For a word u in face labels and a face h, this is the parity, in $\{0, 1\}$, of the number of occurrences of h in u.
$\mathrm{supp}(w)$	The support of a face word w, namely the set of distinct face labels that occur in w.
clique, simplex, flag	A clique in a graph is a set of vertices that are pairwise adjacent. A k-simplex is the combinatorial analogue of a filled k-dimensional triangle; for example, a 1-simplex is an edge and a 2-simplex is a filled triangle. A simplicial complex is flag if every finite clique in its 1-skeleton is filled in as a simplex.

Two faces f, g are called disjoint if their boundary cycles are vertex-disjoint, i.e. $\partial f \cap \partial g = \emptyset$. They meet if their boundary cycles have a common vertex. If a resonance edge admits more than one facial representative with the same boundary-edge set, we fix one representative label once and for all whenever we write a face word. This convention affects only notation, not the rotation itself.

2 Closed rotation words and parity

Let

$$W = M_0 M_1 \cdots M_t$$

be a path in $R(G;\mathcal{F})$, meaning that consecutive matchings M_{i-1} and M_i are adjacent in the resonance graph. A *face word* for W is a sequence

$$w = f_1 f_2 \cdots f_t, \qquad f_i \in \mathcal{F},$$

where juxtaposition denotes a word, not multiplication, such that

$$M_{i-1} \triangle M_i = E(f_i), \qquad i = 1, \ldots, t.$$

If W is closed, i.e. if $M_t = M_0$, then w is called a *closed face word*. For a word u and a face h, let

$$\varepsilon_u(h) \in \{0, 1\}$$

denote the parity of the number of occurrences of h in u.

The following is the closed-walk form of the parity lemma of Tratnik and Ye. We include the proof because we shall apply it to closed walks with repetitions, not only to simple cycles.

Lemma 2.1. *Let $\mathcal{F} \subsetneq F(G)$, and let $w = f_1 \cdots f_t$ be a closed face word in $R(G;\mathcal{F})$. Then every face of G occurs in w an even number of times. In particular, every face of $\mathcal{F}$ that occurs in w occurs at least twice.*

Proof. Let the corresponding closed path be

$$M_0 M_1 \cdots M_t = M_0.$$

Since $M_{i-1} \triangle M_i = E(f_i)$ for each i, we have

$$E(f_1) \triangle E(f_2) \triangle \cdots \triangle E(f_t) = \varnothing.$$

Hence, for each edge e of G, the number of selected facial boundaries containing e is even. If the two face-sides incident with e belong to faces a and b, this gives

$$\varepsilon_w(a) + \varepsilon_w(b) \equiv 0 \pmod 2,$$

and therefore $\varepsilon_w(a) = \varepsilon_w(b)$. By connectedness of the face-dual, all faces of G have the same occurrence parity. Since $\mathcal{F}$ is a proper subset of $F(G)$, some face $q \notin \mathcal{F}$ never occurs in w; thus $\varepsilon_w(q) = 0$. Therefore every face has parity 0. □

The notation $M \xrightarrow{f} N$ used below means that $N = M \triangle E(f)$ and that ∂f is alternating with respect to the matching at the tail of the arrow. Thus a displayed chain of arrows is a path in the resonance graph, with the labels recording the successive faces flipped.

Lemma 2.2. *Let $f, g \in \mathcal{F}$ be disjoint faces. If*

$$M \xrightarrow{f} M \triangle E(f) \xrightarrow{g} M \triangle E(f) \triangle E(g)$$

is a path in $R(G;\mathcal{F})$, then

$$M \xrightarrow{g} M \triangle E(g) \xrightarrow{f} M \triangle E(g) \triangle E(f)$$

is also a path. Thus the two length-two paths bound a square.

Proof. Flipping f changes only the membership of edges of ∂f, and flipping g changes only the membership of edges of ∂g. Since ∂f and ∂g are vertex-disjoint, flipping one of them does not affect the alternating status of the other. Hence the two rotations commute. $\square$

3 The cubical complex associated with a component

Let H be a connected component of $R(G;\mathcal{F})$. We construct a cube complex X_H as follows. Its vertices are the vertices of H, i.e. the perfect matchings in this component, and its edges are the edges of H.

More generally, suppose that $M \in V(H)$ and that

$$S = \{f_1, \dots, f_k\} \subseteq \mathcal{F}$$

is a set of pairwise disjoint faces such that every f_i is M-alternating. The rotations in S can be performed independently. For each $T \subseteq S$, put

$$M_T = M \triangle \left(\triangle_{f\in T} E(f)\right).$$

The 2^k matchings M_T, $T \subseteq S$, span a Boolean k-cube in H: the subsets $T \subseteq S$ play the role of the binary vectors in $\{0,1\}^k$, and two vertices M_T and $M_{T'}$ are adjacent exactly when T and T' differ by one face. Here M_T is the matching obtained from M by flipping exactly the faces in T. We attach one Euclidean k-cube for each such Boolean family of vertices, identifying duplicate descriptions of the same cube. Equivalently, X_H is the cubical complex whose cubes are precisely the Boolean subgraphs obtained from simultaneously flippable, pairwise disjoint facial cycles.

The 1-skeleton $X_H^{(1)}$ is the ordinary graph obtained from X_H by keeping only its vertices and one-dimensional cubes. Thus attaching a square, a three-dimensional cube, or a higher cube records extra homotopies among paths, but it does not create new vertices or new graph edges. In the present construction, the vertices and edges are exactly those of the component H.

For a cube complex X and a vertex $x \in V(X)$, the notation

$$\mathrm{Lk}_X(x)$$

denotes the *link* of x in X. This is a simplicial complex recording the local directions at x. Its vertices are the germs of edges of X issuing from x; here a germ may be read simply as the direction of an edge as it leaves x. Several such link vertices span a simplex precisely when the corresponding directions are contained in one common cube with initial vertex x.

For example, if X is a filled square and x is a corner, then two edges leave x and they lie in the same square; hence $\mathrm{Lk}_X(x)$ consists of two vertices joined by an edge. If the same two edges form only an “L” shape and the square is not filled in, then the link still has two vertices, but they are not joined. If X is a solid three-dimensional cube and x is a corner, then the three coordinate directions from x span a filled triangle in $\mathrm{Lk}_X(x)$, because all three directions lie in one common 3-cube.

In our complex X_H, a vertex is a perfect matching M. An edge germ at M is a possible first flip

$$M \xrightarrow{f} M \triangle E(f).$$

Thus the vertices of $\mathrm{Lk}_{X_H}(M)$ are the facial rotations that are legal at M. A collection of such link vertices spans a simplex exactly when the corresponding facial rotations can be performed simultaneously, equivalently when they generate one cube of X_H. A simplicial complex is *flag* if every finite clique in its 1-skeleton spans a simplex; in other words, pairwise compatibility of directions already implies simultaneous compatibility.

Lemma 3.1. *The 1-skeleton of X_H is H, and every vertex link of X_H is a flag simplicial complex.*

Proof. The 1-skeleton is H by construction. Fix a vertex M of X_H. By the definition of the link, the vertices of $\mathrm{Lk}_{X_H}(M)$ are precisely the edge germs at M in X_H. These edge germs are represented by the facial rotations $f \in \mathcal{F}$ that are M-alternating and whose flip gives an edge of H; here we use the fixed representative labels chosen in the introduction. A set of such link vertices spans a simplex exactly when the corresponding faces are pairwise disjoint. Indeed, if they are pairwise disjoint, Lemma 2.2 shows that their rotations are independent and therefore generate a cube of X_H. Conversely, every cube of X_H was attached from a pairwise disjoint set of facial rotations. Hence a clique in the link means that the rotations are pairwise disjoint; the preceding construction then fills in the whole simplex. Thus every clique in the link spans a simplex, and the link is flag. □

Proposition 3.2. *If X_H is simply connected, then H is a median graph.*

Proof. By Lemma 3.1, X_H satisfies the flag-link condition. If X_H is simply connected, Gromov's criterion implies that X_H is a CAT(0) cube complex [3, Chapter II.5]. The 1-skeleton of a CAT(0) cube complex is a median graph [4]. Since $X_H^{(1)} = H$, the graph H is median. □

It remains to prove that X_H is simply connected.

4 A local parity-propagation lemma

Let f be an even face and let

$$P = N_0 N_1 \cdots N_s$$

be a path in $R(G;\mathcal{F})$ with face word $u = g_1 \cdots g_s$. Assume that f does not occur in u. For a face h, $\varepsilon_u(h)$ denotes the parity of the number of occurrences of h in u; if h does not occur, this parity is 0.

Lemma 4.1. *Assume that f is N_0-alternating and N_s-alternating. Let v be a vertex of ∂f. Then all face sectors at v different from the sector of f have the same occurrence parity in u. Equivalently, all faces of G distinct from f that are incident with v have the same parity in u.*

Proof. List the edges incident with v in cyclic order around v as

$$e_0, e_1, \ldots, e_{d-1},$$

where e_0 and e_{d-1} are the two boundary edges of f, and where the sector between e_{d-1} and e_0 is the sector of f. For $i = 1, \ldots, d-1$, let h_i be the face sector between e_{i-1} and e_i. These are precisely the face sectors at v different from the sector of f. Put

$$\eta_i = \varepsilon_u(h_i),$$

where this is the parity of the face containing the sector h_i. If the same face occurs in more than one sector, the corresponding η_i are the same number.

Each time a face is flipped, it toggles exactly the boundary edges of that face. Therefore, for an edge incident with v, its membership in the matching changes once for each occurrence of one of the two face sectors adjacent to that edge. Counting modulo 2, the parity with which the membership of an incident edge changes from N_0 to N_s is the mod-2 sum of the occurrence parities of the two adjacent face sectors. Since f does not occur in u, the change parities at v are

$$e_0 : \eta_1, \qquad e_i : \eta_i + \eta_{i+1} \pmod 2 \quad (1 \le i \le d-2), \qquad e_{d-1} : \eta_{d-1}.$$

Because f is alternating at v with respect to both N_0 and N_s, exactly one of the two f-boundary edges e_0, e_{d-1} is the matching edge incident with v in each endpoint matching. Hence no edge e_i with $1 \le i \le d-2$ changes membership, while the two edges e_0 and e_{d-1} either both change membership or both do not. Therefore

$$\eta_i = \eta_{i+1} \quad (1 \le i \le d-2), \qquad \eta_1 = \eta_{d-1}.$$

Thus all η_i are equal, as required. □

Corollary 4.2. *With the hypotheses of Lemma 4.1, the parities $\varepsilon_u(h)$ are equal for all faces $h \ne f$ with $\partial h \cap \partial f \ne \varnothing$. Consequently, if one such face has parity zero in u, then every face meeting f has parity zero in u.*

Proof. By Lemma 4.1, for each vertex $v \in V(\partial f)$ there is a local parity c_v such that every face incident with v and distinct from f has parity c_v. If vv' is an edge of ∂f and r is the face on the other side of that boundary edge, then $r \ne f$ and r is incident with both v and v'. This common neighboring face transmits the equality from the local picture at v to the local picture at v', giving

$$c_v = \varepsilon_u(r) = c_{v'}.$$

Since the boundary cycle ∂f is connected, all c_v are equal. The conclusion follows. □

5 Reduced closed words

We now translate homotopies in the 2-skeleton of X_H into operations on face words. The 2-skeleton is the subcomplex consisting of all vertices, all edges, and all square 2-faces of X_H. Higher-dimensional cubes do not add new fundamental-group relations beyond those already visible in their square faces, so, by cellular approximation, simple connectedness of X_H is equivalent to null-homotopy of all closed edge paths in $X_H^{(1)} = H$.

There are two elementary moves on face words:

(i) deletion of a consecutive pair ff. This means that the path flips f and then immediately flips f back, so the two steps are an immediate backtrack along the same resonance edge;

(ii) interchange of consecutive letters $fg \leftrightarrow gf$ when $\partial f \cap \partial g = \varnothing$. This is the square homotopy from Lemma 2.2: the two disjoint flips can be performed in either order and lead to the same matching.

These moves preserve the homotopy class of the corresponding loop in X_H.

A nonempty closed face word is read cyclically, because a closed loop has no distinguished starting edge. Thus a cyclic conjugate of $f_1 f_2 \cdots f_m$ is obtained by choosing a different starting point, for instance $f_i f_{i+1} \cdots f_m f_1 \cdots f_{i-1}$. A closed face word is called *fully reduced* if no cyclic conjugate of any word obtained from it by a sequence of disjoint commutations contains a consecutive pair ff. A shortest non-null-homotopic edge loop in X_H is necessarily fully reduced: otherwise square homotopies followed by deletion of a backtrack would produce a shorter loop in the same nontrivial homotopy class.

Lemma 5.1. *Let w be a fully reduced closed face word. Suppose that, cyclically,*

$$w = fufu'$$

where u' is another, possibly empty, word and u contains no occurrence of f. Then some face occurring in u meets f.

Proof. If every face occurring in u were disjoint from f, then repeated use of Lemma 2.2 would commute the first displayed occurrence of f across all letters of u, producing a cyclic word with a consecutive pair ff. This contradicts full reducedness. $\square$

Theorem 5.2. *For every connected component H of $R(G;\mathcal{F})$, the cubical complex X_H is simply connected.*

Proof. Suppose, for a contradiction, that some closed edge path in $H = X_H^{(1)}$ is not null-homotopic in X_H. Choose such a path of minimum length, and let

$$w = f_1 f_2 \cdots f_m$$

be a face word for it. As observed above, w is fully reduced. Put

$$A = \{f_i : 1 \leq i \leq m\}.$$

By Lemma 2.1, each face in A occurs an even number of times. Moreover $A \subseteq \mathcal{F} \subsetneq F(G)$, so A is a proper subset of the full face set. Since the face-dual is connected and A is nonempty, there are faces $p_0 \in A$ and $p_{-1} \notin A$ such that p_{-1} shares a boundary edge with p_0.

Choose two consecutive cyclic occurrences of p_0 in w and write the corresponding cyclic subword as

$$p_0 U_0 p_0,$$

where U_0 contains no occurrence of p_0. The face p_0 is alternating at both ends of the path segment represented by U_0: immediately after a flip of p_0 it is still alternating, with the complementary set of boundary edges selected, and immediately before the second displayed occurrence it is alternating because that second flip is legal. The face p_{-1} does not occur in U_0, since $p_{-1} \notin A$.

We shall construct inductively faces

$$p_0, p_1, p_2, \dots$$

and nonempty words

$$U_0 \supsetneq U_1 \supsetneq U_2 \supsetneq \cdots,$$

where the symbol $\supsetneq$ means proper containment as contiguous subwords: each U_{i+1} is a proper contiguous subword of U_i. Suppose that p_i and U_i have been constructed with the following properties:

(a) p_{i-1} meets p_i;

(b) p_{i-1} does not occur in U_i;

(c) $p_i U_i p_i$ is formed by two consecutive occurrences of p_i inside the previous word (inside the cyclic word w when $i = 0$, and inside U_{i-1} when $i \geq 1$); in particular p_i does not occur in U_i;

(d) p_i is alternating at both ends of the path segment represented by U_i.

The initial pair (p_0, U_0) satisfies these properties, with p_{-1} as above.

Apply Corollary 4.2 to the face p_i and the word U_i. Since p_{i-1} meets p_i and has parity zero in U_i, every face meeting p_i has parity zero in U_i. By Lemma 5.1, the word U_i contains at least one face meeting p_i; choose one and call it p_{i+1}. Its occurrence parity in U_i is zero and it occurs at least once, so it occurs at least twice. Choose two consecutive occurrences of p_{i+1} inside U_i and write the corresponding subword as

$$p_{i+1} U_{i+1} p_{i+1}.$$

Then U_{i+1} contains no occurrence of p_{i+1} and is a proper contiguous subword of U_i. It cannot be empty, for otherwise the original fully reduced cyclic word w would contain a consecutive pair $p_{i+1} p_{i+1}$. Finally, p_i does not occur in U_{i+1} because U_i contains no occurrence of p_i. The face p_{i+1} is alternating at both ends of the path segment represented by U_{i+1}: it is alternating immediately after the first displayed flip of p_{i+1}, and it is alternating immediately before the second displayed flip because that flip is legal.

Thus the induction continues forever and produces an infinite strictly descending chain of nonempty finite words. This is impossible because their lengths strictly decrease. Hence no non-null-homotopic closed edge path exists, and X_H is simply connected. □

6 Proof of the main theorem and final remarks

Proof of Theorem 1.1. Let H be a connected component of $R(G; \mathcal{F})$. If H consists of a single vertex, then it is a median graph. Otherwise construct X_H as in Section 3. By Lemma 3.1, all vertex links of X_H are flag. By Theorem 5.2, X_H is simply connected. Therefore X_H is a CAT(0) cube complex by Gromov's criterion, and its 1-skeleton is a median graph. Since $X_H^{(1)} = H$, the component H is median. □

Remark 6.1. *The assumption $\mathcal{F} \subsetneq F(G)$ is used in two essential places. First, it gives the parity-of-labels statement for closed face words, Lemma 2.1. Second, for the support A of a putative minimal non-null-homotopic loop, it guarantees a face outside A; connectedness of the face-dual then supplies the initial adjacent pair $p_{-1} \notin A$ and $p_0 \in A$ that starts the descending-word argument. This also explains why the case $\mathcal{F} = F(G)$ is genuinely different.*

Remark 6.2. *The proof does not use the particular induced-hypercube embedding constructed in [6]. The cubical complex X_H is determined intrinsically by independent facial rotations. Medianity follows from nonpositive curvature of this complex, not from an a priori isometric embedding of H into a hypercube.*

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